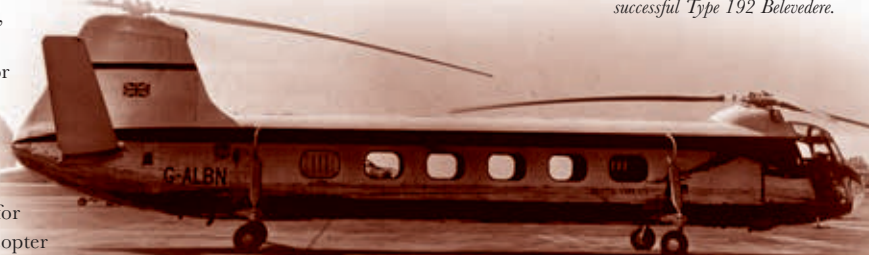


EARLY HELICOPTERS

GERMAN PLANEMAKER HEINRICH FOCKE, a founder of the Focke-Wulf company, led the world in helicopter development in the late 1930s. Focke-designed helicopters were ordered into production for military use during WWII, but few entered service. Meanwhile, Igor Sikorsky's experimental 1939 single-rotor VS-300 led to the R-4 Hoverfly, which went into production for the US Army in 1942. Unlike its German equivalents, the R-4 had a chance to prove its usefulness in combat. The Bell company also began helicopter development during the war, and in 1946 the highly successful Bell 47 became the first helicopter licensed for civilian use. By the early 1950s, some 30 types of helicopter were flying, including: variants on the classic Sikorsky design with a single horizontal rotor and vertical tail rotor; machines with two counter-rotating horizontal rotors; and helicopters with small jets at the tips of the rotor blades, which obviated the need for a tail rotor.



BELVEDERE ANCESTOR

The prototype Type 173 first flew in 1952 and developed into the RAF's successful Type 192 Belvedere.

Cierva C.8 Mk.IV Autogiro

Rotary-winged flight was achieved long before the first practical helicopter appeared. In 1920, Juan de la Cierva began experimenting with this form of aircraft which he called an "Autogiro." It had an unpowered rotor, so it could not take off vertically or fly backwards, but it came close to a vertical landing. On September 18, 1928, a C.8L made the first crossing of the English Channel by a rotating wing aircraft.



ENGINE 200hp Armstrong Siddeley Lynx IVC 7-cylinder radial

ROTOR SPAN 39ft 8in (12.1m) **LENGTH** 28ft 6in (8.7m)

TOP SPEED 100mph (161kph) **CREW** 1

PASSENGERS 1

Fairey Jet Gyrodyne



The original Gyrodyne first flew in December 1947, combining autogyro and helicopter features. In 1948 it set a new world rotorcraft speed record of 124mph (200kph). The Jet Gyrodyne appeared in January 1954, powered by compressed air from the engine fed to jet units at the tips of the two-blade rotor.

ENGINE 525hp Alvis Leonides 9-cylinder radial

ROTOR SPAN 60ft (18.3m) **LENGTH** 25ft (7.6m)

TOP SPEED 140mph (225kph) **CREW** 2

PASSENGERS 1

Focke-Achgelis Fa 61

The Fa 61 was the first fully controllable helicopter and flew in June 1936. Although influenced by the Cierva autogiro, the new machine used twin rotors which gave remarkable agility. After establishing a number of world rotorcraft records, the aircraft was flown indoors in the Berlin Deutschlandhalle in February 1938 by the brilliant pilot, Hanna Reitsch, who demonstrated its ease of control and manoeuvrability to the crowd.

ENGINE 160hp Bramo Sh.14a 7-cylinder radial

ROTOR SPAN 23ft (7m) **LENGTH** 24ft (7.3m)

TOP SPEED 62mph (100kph) **CREW** 1

PASSENGERS None



Piasecki YH-16A Transporter



Together with Sikorsky, Frank Piasecki was a pioneer of American helicopter design. From the mid-1940s, his company built a series of twin-rotor machines from which the current Boeing Chinook is a direct descendant. An improved version of the 1953 prototype, with shaft turbines replacing the piston engines, first flew in 1955. A year later the project was canceled.

ENGINE 2 x 1800shp Allison YT-38-10 turboshaft

ROTOR SPAN 82ft (25m) **LENGTH** 77ft 7in (23.8m)

TOP SPEED 146mph (235kph) **CREW** 3

PASSENGERS 40 troops

Sikorsky VS-300

A young Igor Sikorsky built an unsuccessful helicopter in 1909, before turning his attention to giant airplanes and flying boats. Thirty years later, he returned to his first love, which would make his name synonymous with helicopters. His VS-300, which made a tethered flight on September 13, 1939, was the first successful American-built helicopter.



ENGINE 75hp 4-cylinder Lycoming (initial tethered flight)

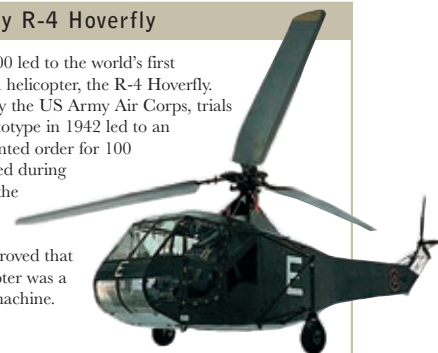
ROTOR SPAN 28ft (8.5m) **LENGTH** 66ft (20.1m)

TOP SPEED 64mph (103kph) **CREW** 1

PASSENGERS None

Sikorsky R-4 Hoverfly

The VS-300 led to the world's first production helicopter, the R-4 Hoverfly. Ordered by the US Army Air Corps, trials of the prototype in 1942 led to an unprecedented order for 100 R-4Bs. Used during WWII by the Allied forces, the Hoverfly proved that the helicopter was a practical machine.



ENGINE 180hp Warner R-550 Super Scarab radial

ROTOR SPAN 38ft (11.6m) **LENGTH** 35ft 3in (10.7m)

TOP SPEED 75mph (121kph) **CREW** 2

PASSENGERS None